

MENG 4205.002

LAB 4: FLUID FRICTION APPARATUS

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ABSTRACT

This experiment examined fluid flow behavior and head losses in pipe bends using the TecQuipment H408 Fluid Friction Apparatus. Straight smooth and rough pipes had been tested in previous experiments to show baseline frictional losses, but the focus of this study was on bend geometries, including long-radius bends of 50 mm, 100 mm, and 150 mm, a mitre bend, and a standard elbow. Pressure differences were measured with piezometer tubes and volumetric flow rates were recorded using the hydraulic bench. From these values, velocities, Reynolds numbers, and bend loss coefficients were calculated. The results confirmed that head loss increases as bend radius decreases. The 150 mm bend produced the lowest loss coefficients where $k \approx 12-13$ while the elbow and mitre fittings produced scattered values ranging from 5-66. These outcomes matched theoretical expectations where sharp bends generate separation and turbulence unlike smoother bends where larger-radius bends redirect flow more efficiently. Deviations from theory can likely be linked to turbulence, entrained air, manometer sensitivity, and in our case skewed flow rates. Despite these discrepancies, the observed trends supported established fluid mechanics principles and highlighted the engineering importance of bend geometry in minimizing energy losses [1][2].

Keywords: Bend losses, Head loss, Loss coefficient, Elbow, Mitre bend, Flow separation, Hydraulic bench

Nomenclature

Q Volumetric Flow Rate (m^3/s)
 h Head loss (m)
 u Velocity (m/s)
 d Pipe diameter (m)
 ν Kinematic Viscosity (m^2/s)
 l Length of pipe between tappings (m)
 f Friction factor (-)
 g Acceleration due to Gravity (m/s^2)
 k Loss coefficient (-)
 Re Reynolds number (-)

A Cross-sectional area (m^2)

1. INTRODUCTION

In practical piping systems, energy is lost not only through wall friction in straight pipes but also in localized regions where the flow is forced to change direction. Among the most common causes of these additional or “minor” losses are bends, elbows, and mitre joints. Unlike straight pipes, which generally exhibit predictable frictional losses, bends introduce complex flow patterns, including curvature-induced secondary motion, flow separation, and localized turbulence. These effects increase the resistance to flow, requiring higher pumping power and impacting overall system efficiency [2][3].

To quantify these effects, minor losses are expressed using a dimensionless loss coefficient, k , which relates head loss to the square of the velocity head. The loss coefficient depends strongly on geometry with long-radius bends producing relatively small k values and sharp elbows or mitres yielding much larger values [2][4]. In engineering practice, this makes bend geometry a critical design consideration. For example, long-radius bends are preferred in high-flow systems such as HVAC ducts, process piping, and cooling circuits, where minimizing head loss reduces both energy consumption and operational cost.

The TecQuipment H408 Fluid Friction Apparatus provides a controlled setting for studying these phenomena. Using piezometer tubes attached to upstream and downstream tapping points, the head loss across a bend can be directly measured. The hydraulic bench enables volumetric measurement of flow, allowing the calculation of flow velocity and Reynolds number [1].

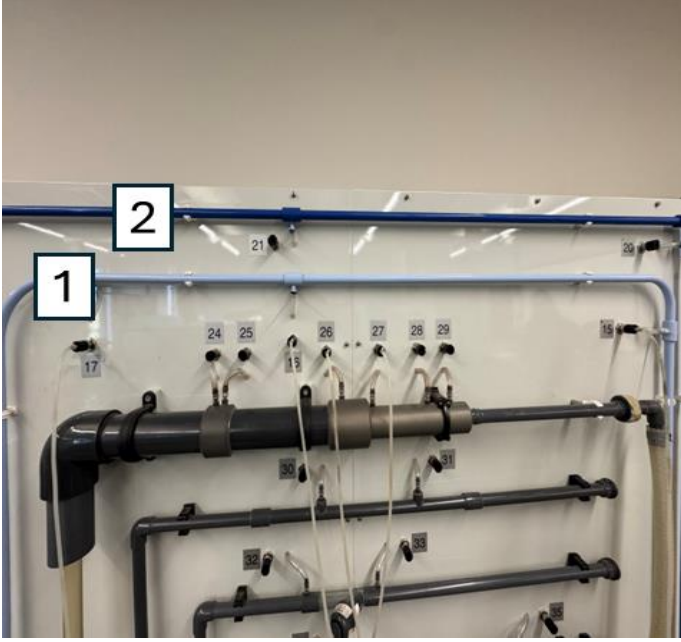


Figure 1 Image shows (1) 100mm bend and (2) mitre bend

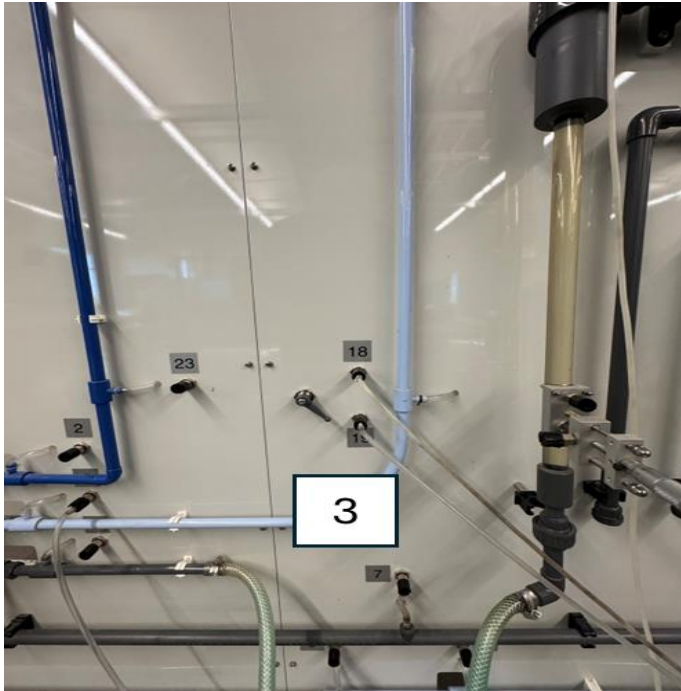


Figure 2 Image shows (3) 50mm bend

In a previous laboratory session, straight smooth and rough pipes were studied to establish baseline frictional losses. Those results confirmed the applicability of the Blasius correlation for smooth pipes and the Moody chart for roughened surfaces [4]. The present investigation builds on that foundation but shifts the focus to bend losses. Specifically, long-radius bends of 50 mm, 100 mm, and 150 mm were compared with sharper geometries including a mitre bend and a standard elbow. The primary objective was to determine how bend radius relative to pipe

diameter (R/d) affects head loss, to compare experimental coefficients with theory, and to evaluate the engineering implications of bend geometry in piping system design [2][3].

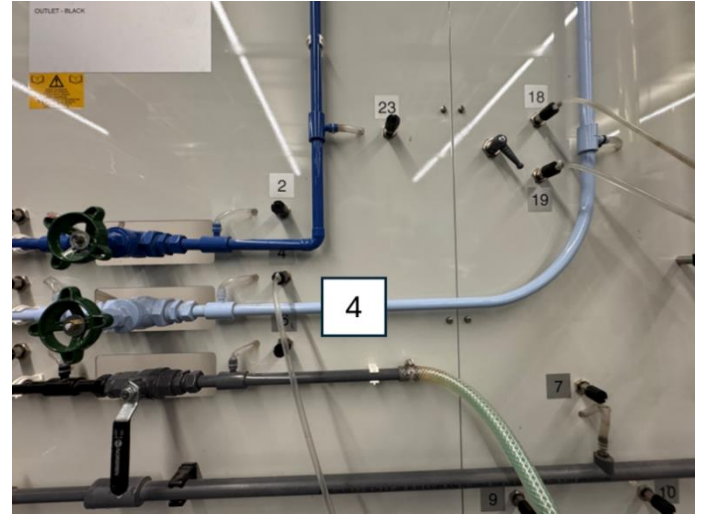


Figure 3 (4) Flow input for 50, 100 and 150mm bend.

2. THEORY

When fluid flows through a pipe bend, additional energy is dissipated beyond the frictional losses of a straight pipe. This dissipation arises from curvature-induced secondary flows, flow separation along the inner and outer walls, and localized turbulence that develops as the flow redirects [2]. The head loss associated with bends is expressed as:

$$h = k(u^2 / 2g) \quad (1)$$

where h is head loss, k is the bend loss coefficient, u is mean velocity, and g is gravitational acceleration.

Because pressure taps are located upstream and downstream of each bend, the measured loss includes both the bend geometry effect and the frictional loss of the straight pipe length between the tapping's. To isolate the geometric contribution, the theoretical straight-pipe friction loss can be estimated using the Darcy–Weisbach equation [2][3]:

$$h_f = f \frac{L}{d} \frac{u^2}{2g} \quad (2)$$

where f is the friction factor, L is the equivalent pipe length, and d is the pipe diameter. For smooth turbulent flow, the Blasius correlation provides an empirical relation [4]

For turbulent flow in smooth pipes, the Blasius correlation provides an empirical expression for the friction factor [4]:

$$f = 0.079 Re^{-1/4} \quad (3)$$

where Reynolds number is given by

$$Re = \frac{ud}{\nu} \quad (4)$$

with ν as the kinematic viscosity of water.

The geometry of the bend itself is typically characterized by the ratio R/d , where R is the radius of curvature of the bend. Larger R/d ratios correspond to smoother bends, which minimize separation and reduce the loss coefficient k . Conversely, small R/d ratios, such as in sharp elbows or mitre joints, generate stronger separation and secondary vortices leading to higher values of k [2][3].

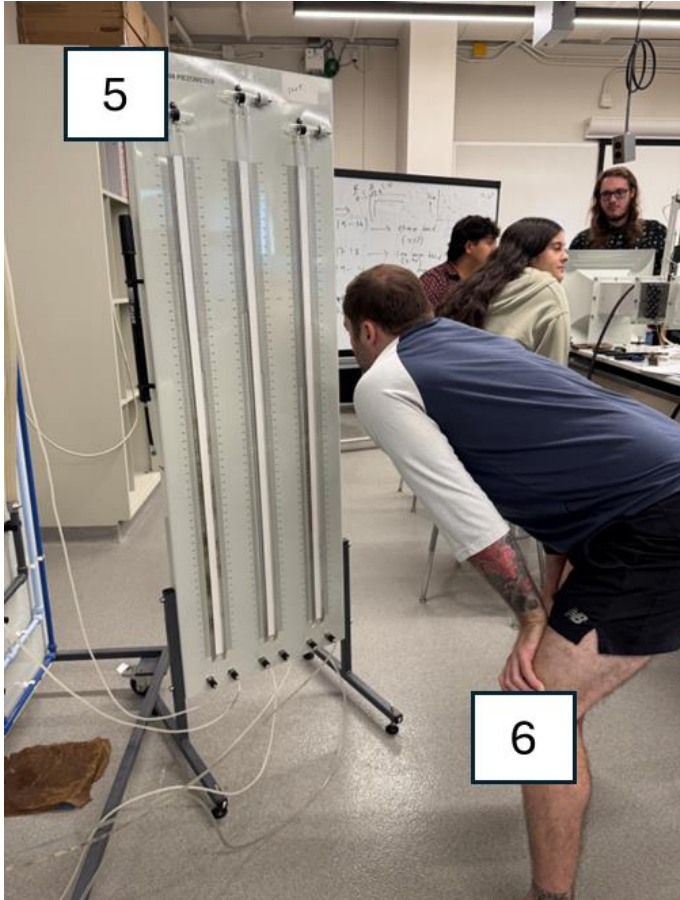


Figure 4 Chase (6) reading (5) piezometer results

In summary, theory predicts that head loss in bends varies with the square of velocity, friction factors depending on Reynolds number, and bend loss coefficients depending strongly on the geometry of the bend. These relationships provide the framework for analyzing the experimental data collected in this study.

3. METHODS

The H408 Fluid Friction Apparatus was mounted on a hydraulic bench, which provided a recirculating water supply. Piezometer tubes were connected to upstream and downstream tapping points for each bend section, allowing head loss to be determined directly from water column differences [1].

Preparation of Apparatus

Before testing, the hydraulic bench was filled and inspected to ensure steady flow. All flexible hose connections were tightened, and piezometer tubes were checked for trapped air. The hand pump provided with the apparatus was used to equalize piezometer levels to mid-scale, ensuring accurate differential readings.

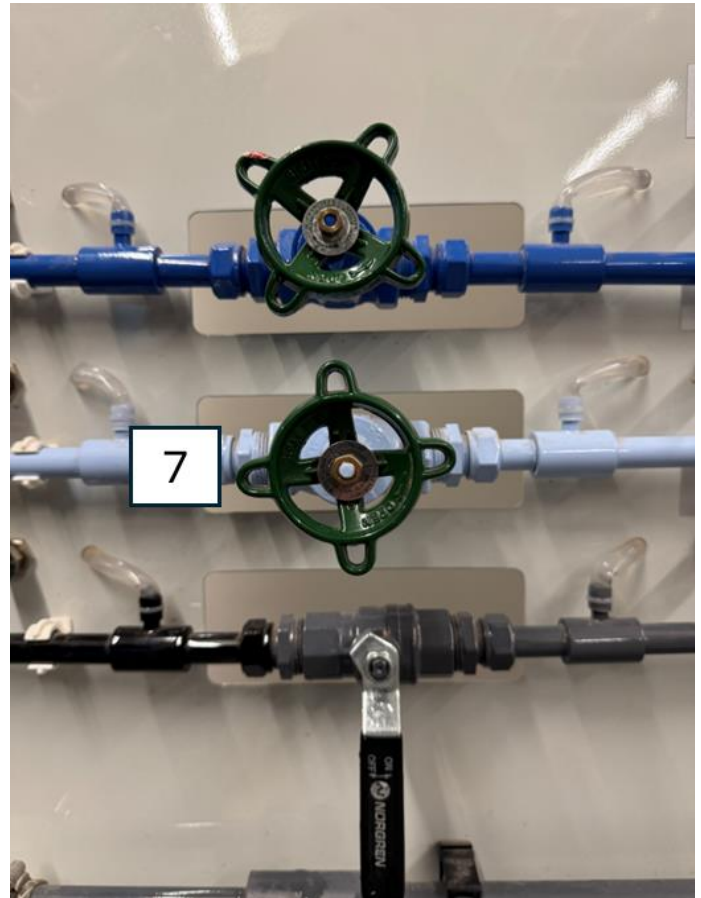


Figure 5 Flow control knob for 50, 100, 150 mm bends (7)

Bleeding and Calibration

Because air bubbles are a major source of error in manometric devices, the system was carefully bled prior to testing. Bleed valves were opened during high flow to flush air from the pressure lines, and piezometer menisci were tapped lightly to stabilize their levels. Flow meters were inspected to confirm no blockages, although they were not central to this study.

3.1 Preparation of Apparatus

Before testing, the hydraulic bench water supply was inspected to ensure a clean, constant flow. All flexible hose connections were tightened, and the piezometer unit was checked for trapped air. Each experimental section, smooth pipe, roughened pipe, bend, valve, or flow meter, was identified, and the associated tapping's related to transparent tubing. The hand pump supplied with the apparatus was used to equalize piezometer levels prior to data collection.

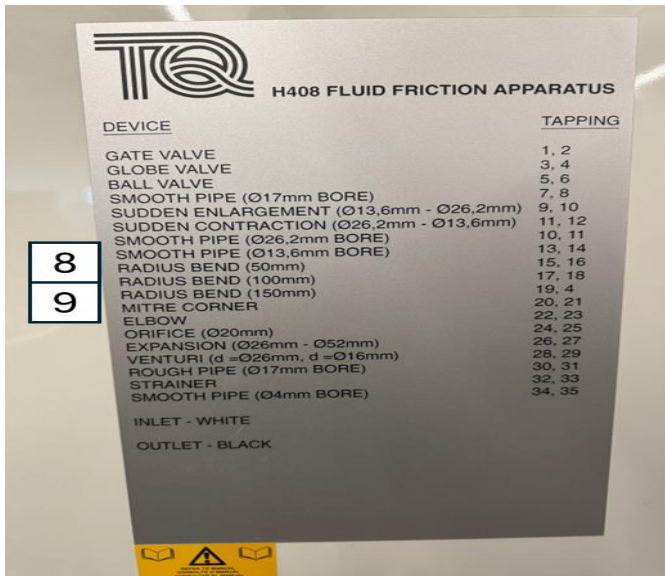


Figure 6 (8) (9) show the path configurations tested

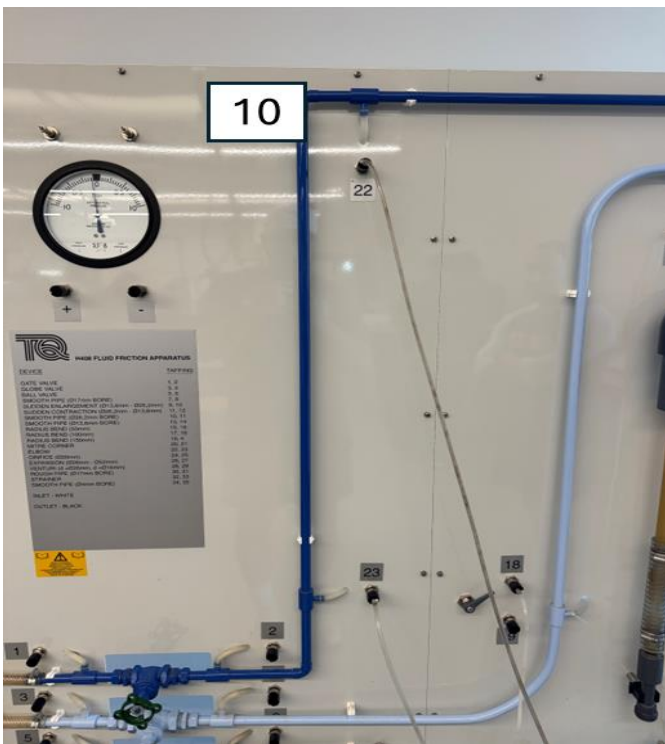


Figure 7 (10) shows the mitre bend tested

3.2 Bleeding and Calibration

Air was bled from all pipes, piezometers, and differential gauges to prevent inaccurate reading. The pressure gauge bleed valves were opened during high flow to flush out air, and the manometer levels were adjusted to mid-scale using the hand pump. Flow meters (orifice and venturi) were inspected to ensure no blockages.

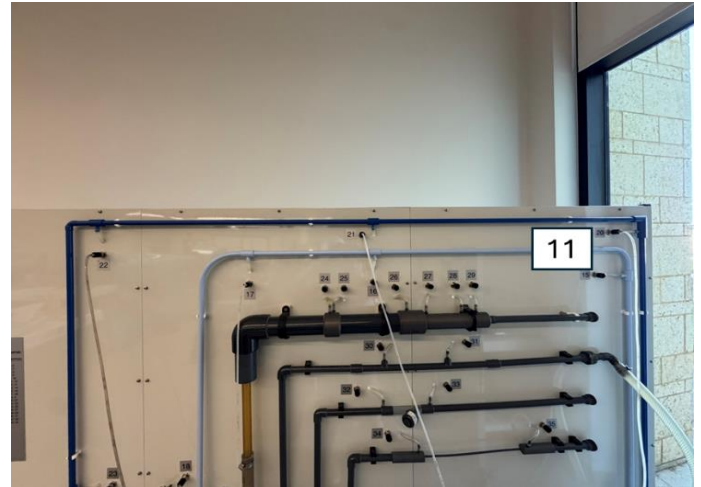


Figure 8 (11) shows full mitre bend flow and direction.

3.3 Data Collection Procedure

Flow Establishment: The hydraulic bench supply valve was opened gradually, and the circuit valve for the selected bend was adjusted while other circuits remained closed.

Baseline Recording: Piezometer readings were taken once the flow stabilized.

Flow Measurement: The volumetric flow rate was determined by timing the collection of a fixed volume, 15L, in the

hydraulic bench reservoir:

$$Q = \frac{L}{A}$$

where V is the collected volume and t is the measured time.

Velocity and Reynolds Number: The mean velocity was calculated as

$$v = \frac{Q}{A}, A = \pi r^2$$

and the Reynolds number as

$$Re = \frac{ud}{\nu}$$

with $\nu = 1.004 \times 10^{-6} \text{ m}^2/\text{s}$ for water.

Head Loss and Loss Coefficient: Head loss was obtained from piezometer differences. The bend loss coefficient was then determined as

$$k_l = h_b / (v^2 / 2g)$$

Repetition Across Flow Conditions: The above steps were repeated for five to six different flow rates per bend geometry to evaluate performance across a range of Reynolds numbers.

At the end of testing, valves were closed gradually to avoid water hammer, the piezometer tubes were drained, and the apparatus was wiped down to prevent corrosion.

3.4 Shut Down

At the end of testing, the control valves were closed gradually to prevent water hammer. The piezometer tubes were allowed to drain back, and all hoses were disconnected. The apparatus was wiped down and left to dry to prevent corrosion.

4. EXPERIMENTS

This laboratory investigation concentrated primarily on head losses in pipe bends. Tests were conducted on three long-radius bends with radii of 50 mm, 100 mm, and 150 mm, as well as a sharp mitre bend and a standard elbow. Each test section contained piezometer tapings upstream and downstream of the bend, allowing head loss to be determined directly from differential water column heights. Flow rates were varied systematically to evaluate the effect of velocity and Reynolds number on measured losses.

Although the apparatus also includes straight smooth and rough pipe sections, valve fittings, and flow measurement devices, these were the focus of previous/later laboratory sessions. The straight pipe results in particular provided useful baseline frictional data, validating the Darcy–Weisbach and Blasius predictions and serving as a reference for separating geometric bend losses from frictional contributions [4]. Their discussion is not repeated here but is cited for context.

In each bending test, water was circulated through the selected circuit while other valves were closed to ensure isolated flow. The volumetric flow rate was measured by timing the collection of a fixed water volume in the hydraulic bench reservoir, while piezometer readings were recorded simultaneously. By repeating this process across a range of flows, data was collected for calculating head loss, Reynolds number, and bend loss coefficients.

By comparing long-radius bends to sharper geometries such as the elbow and mitre, the experiment enabled direct evaluation of how bend radius relative to pipe diameter (R/d) influences head loss. This comparison also highlighted the engineering trade-off between compact layouts, which introduce higher energy losses, and smooth-radius bends which improved efficiency but require more space.

5. RESULTS AND DISCUSSION



Figure 9 (12) shows an error that affected our fill rate in the tub.

5.1 Results

The raw data collected included flow collection times, piezometer readings, and water column differences across each bend. From these, volumetric flow rates, average velocities, Reynolds numbers, and head losses were calculated. The bend loss coefficient, k , was then determined.

Representative results showed that bend geometry had a significant influence on head loss. The 150 mm radius bend, $R/d \approx 11$, produced the lowest losses with k values around 12–13. The 100 mm and 50 mm radius bends showed higher losses with coefficients near 13–20 at moderate Reynolds numbers. The mitre bend and elbow generated the highest and most scattered results with k ranging between 5 and 66 depending on the flow rate.

Plots of head loss versus velocity squared yielded near-linear trends, consistent with the theoretical dependence [$h \propto u^2$]. At low flow rates deviations were observed due to piezometer instability. However, at higher flows turbulence and flow separation introduced additional scatter.

Straight-pipe friction tests from the previous lab session were used for comparison. Smooth-pipe data generally followed the Blasius correlation, while rough-pipe data aligned with the Moody chart within experimental error [4]. These baselines confirmed that the large additional losses observed here were attributable to bend geometry rather than wall friction alone.

Tables summarizing calculated results and figures of the performance curves are included to highlight these relationships and allow comparison to theoretical predictions.

Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
5.8E-01	1.2E+04	7.6E-03	5.9E-03	0.34	0.017	20.57
5.5E-01	1.1E+04	7.7E-03	5.4E-03	0.31	0.015	20.25
4.8E-01	9.6E+03	8.0E-03	4.2E-03	0.24	0.012	21.16
2.0E-01	4.0E+03	9.9E-03	9.2E-04	0.12	0.002	58.36
7.5E-02	1.5E+03	1.3E-02	1.6E-04	0.01	0.000	42.25

Table 1 50 mm radius bend experimental results

Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
5.8E-01	1.2E+04	7.6E-03	5.5E-03	0.21	0.017	12.69
5.5E-01	1.1E+04	7.7E-03	5.1E-03	0.19	0.015	12.36
4.8E-01	9.6E+03	8.0E-03	4.0E-03	0.14	0.012	12.21
2.0E-01	4.0E+03	9.9E-03	8.7E-04	0.08	0.002	38.74
7.5E-02	1.5E+03	1.3E-02	1.5E-04	0.01	0.000	52.81

Table 2 100 mm radius bend experimental results

Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
5.8E-01	1.2E+04	7.6E-03	4.2E-03	0.21	0.017	12.63
5.5E-01	1.1E+04	7.7E-03	3.8E-03	0.19	0.015	12.68
4.8E-01	9.6E+03	8.0E-03	3.0E-03	0.15	0.012	13.07
2.0E-01	4.0E+03	9.9E-03	6.5E-04	0.07	0.002	36.29
7.5E-02	1.5E+03	1.3E-02	1.2E-04	0.01	0.000	52.81

Table 3 150 mm radius bend experimental results

Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
1.2E+00	1.6E+04	7.0E-03	3.6E-02	0.39	0.073	5.784
1.1E+00	1.5E+04	7.1E-03	3.2E-02	0.33	0.064	5.7
6.1E-01	8.3E+03	8.3E-03	1.1E-02	0.28	0.019	15.01
4.4E-01	6.0E+03	9.0E-03	6.2E-03	0.15	0.010	15.32
9.0E-02	1.2E+03	1.3E-02	3.8E-04	0.02	0.000	55.97

Table 4 Mitre corner experimental results

Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
1.2E+00	1.6E+04	7.0E-03	3.6E-02	0.37	0.073	5.539
1.1E+00	1.5E+04	7.1E-03	3.2E-02	0.32	0.064	5.514
6.1E-01	8.3E+03	8.3E-03	1.1E-02	0.27	0.019	14.54
4.4E-01	6.0E+03	9.0E-03	6.2E-03	0.14	0.010	15.01
9.0E-02	1.2E+03	1.3E-02	3.8E-04	0.03	0.000	65.71

Table 5 Elbow bend experimental results

5.2 Sample Calculations

An example calculation is shown for the 150mm bend with a flow rate of 98 seconds in 15 liters

Volumetric Flow Rate

$$Q = \frac{0.15}{98} = 1.53(10^{-4}) \frac{m^3}{s}$$

Average Velocity

$$u = \frac{Q}{A} = \frac{1.53(10^{-4})}{3.205(10^{-4})} = 0.48 \frac{m}{s}$$

Reynolds Number

$$Re = \frac{ud}{\nu} = \frac{0.51(0.0202)}{1.004(10^{-6})} = 9609.27$$

Head Loss from Piezometer Readings

$$h = \frac{\Delta p}{\rho g} = \Delta h = 0.15m$$

Friction Factor

$$k_L = \Delta h \div \left(\frac{u^2}{2g} \right) = 0.22 \div \left(\frac{0.48^2}{19.62} \right) = 12.82$$

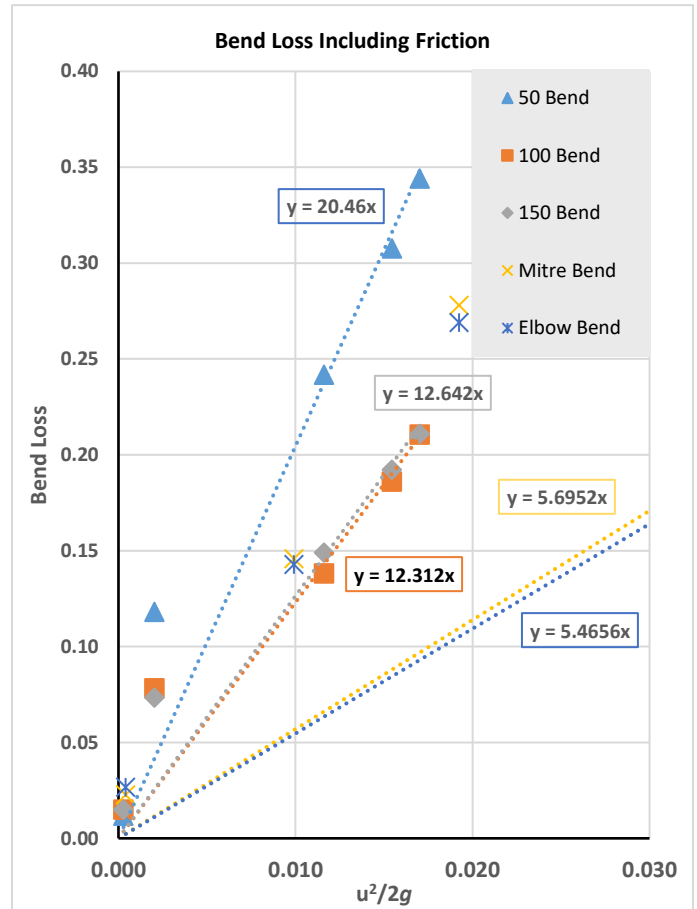


Figure 10 Bend head loss versus velocity head ($v^2/2g$). The slope of each line represents the bend loss coefficient k_B

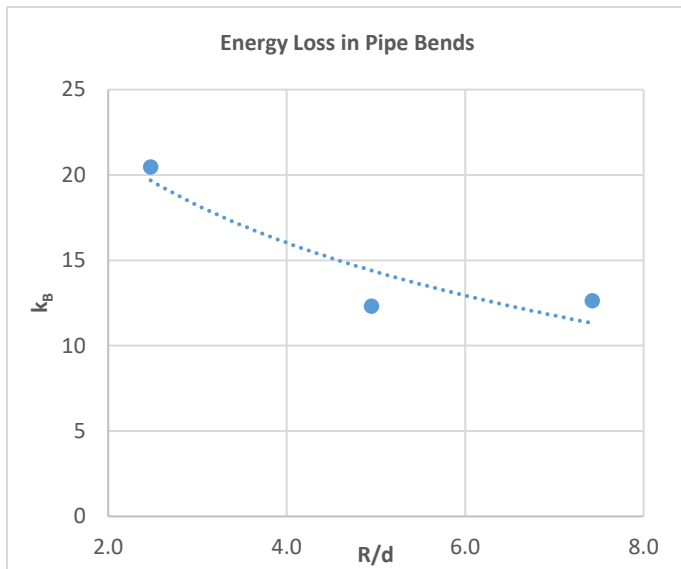


Figure 11 Bend loss coefficient (k_B) as a function of bend radius to diameter ratio (R/d), showing the effect of bend geometry on energy losses

5.3 Ideal vs. Actual Comparison

Theory predicts that larger-radius bends should have smaller loss coefficients, while sharp elbows and mitres should produce higher values due to separation and turbulence. Our results generally followed this trend. The 150 mm bend gave the lowest k which is close to expected value unlike the 100 mm and 50 mm bends which were higher. The elbow and mitre showed the largest and most scattered results.

Some differences appeared at low flow rates where piezometer readings were less stable at higher flows and when turbulence introduced scatter. Even with these deviations, the results confirmed that bend radius has a strong influence on energy losses.

5.4 Discussion

If you were designing a piping system with 13.6 mm inside diameter pipe, and wanted to reduce the losses due to bends, what would you set as the minimum bend radius?

For a pipe with an inside diameter of 13.6 mm a bend radius at least two to three times the diameter would minimize losses.

That means the minimum bend radius should be around 27–41 mm. In practice, using a larger bend radius than this range would further reduce separation and turbulence, but 2–3D is often the accepted minimum to balance space and efficiency.

Why do standard graphs of k_L against R/d show that k_L has a minimum value at R/d between 2 and 3?

When the bend radius is very small, R/d approaches 0, it flows separately strongly therefore creating high losses. As the radius increases, the flow follows the bend more smoothly and losses drop. However, once R/d gets larger than about 3, the additional length of pipe adds extra wall friction, so the total loss k_L starts to increase again. This trade-off explains why the

minimum occurs between 2 and 3. Our experimental data (Figure 11) captured only the decreasing portion of this relationship, as all tested bend radii ($R/d = 2.5$ to 7.4) fell at or above the optimal range where the minimum occurs

Which value, k_B or k_L , is of most practical use and why?

In practice, k_L is more useful because it represents the total loss around a bend, including both the bend geometry and the straight-pipe friction of the bend length. Engineers need this total loss when sizing pumps and estimating energy consumption in real systems. While k_B is helpful for separating the geometric effect in experiments, k_L gives the values actually needed in design.

6 CONCLUSION

The bending experiments with the TecQuipment H408 Fluid Friction Apparatus showed that head losses in bends increase as the bend radius gets smaller. The long-radius bend (150 mm) had the lowest loss coefficients, around $k \approx 12$ –13, which confirmed that smoother more gradual bends reduce separation and energy loss. The tighter 100 mm and 50 mm bends had much higher values $k \approx 12$ –20, while the mitre bend and elbow produced the largest and most scattered readings $k \approx 5$ –66 due to stronger turbulence and secondary flows.

Even though our results followed the general trend predicted by fluid mechanics theory, there were some major deviations. At low flow rates, the k values were unrealistically high. This happened because of both real flow behavior and several experimental problems. Our flow rates were very low during testing, which caused flow separation and unstable transitions inside the bends. The small velocity head ($v^2/2g$) also made any pressure loss look much larger when calculating k . On top of that, our volumetric tank was leaking during timing, which threw off our flow-rate calculations, especially at lower flows. The piezometers kept fluctuating because of air bubbles and pressure changes from the leaks and there were additional small leaks in the pipe system that caused extra pressure drops. Altogether, these issues made the lowest-flow data unreliable and gave k values much higher than those listed in the equipment manual.

From an engineering standpoint, the experiment showed why bend geometry and accurate measurement are both important. Long-radius bends take up more space but are much more efficient and reduce energy loss. Sharp elbows and mitres are more compact but cause higher losses and lower system efficiency. This lab helped show how design choices, proper setup, and steady flow conditions all play a big role in achieving accurate results and better system performance.

REFERENCES

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APPENDIX

Pipe type: 50 mm Radius Bend
 Internal Diameter (d) = 0.0202 m
 Area (A) = 0.0003205 m²
 Length (l) = 0.92 m

Time for 15 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Upstream Tapping (mm)	Downstream Tapping (mm)	Difference (Δh) (m)	Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
81	1.85E-04	1076	726	0.35	0.58	1.2E+04	0.0076	0.0059	0.34	0.02	20.57
85	1.76E-04	1045	732	0.31	0.55	1.1E+04	0.0077	0.0054	0.31	0.02	20.25
98	1.53E-04	988	742	0.25	0.48	9.6E+03	0.0080	0.0042	0.24	0.01	21.16
234	6.41E-05	876	757	0.12	0.20	4.0E+03	0.0099	0.0009	0.12	0.00	58.36
627	2.39E-05	781	769	0.01	0.07	1.5E+03	0.0127	0.0002	0.01	0.00	42.25

Pipe type: 100 mm Radius Bend
 Internal Diameter (d) = 0.0202 m
 Area (A) = 0.0003205 m²
 Length (l) = 0.864 m

Time for 15 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Upstream Tapping (mm)	Downstream Tapping (mm)	Difference (Δh) (m)	Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
81	1.85E-04	943	727	0.22	0.58	1.2E+04	0.0076	0.0055	0.21	0.02	12.69
85	1.76E-04	919	728	0.19	0.55	1.1E+04	0.0077	0.0051	0.19	0.02	12.36
98	1.53E-04	872	730	0.14	0.48	9.6E+03	0.0080	0.0040	0.14	0.01	12.21
234	6.41E-05	792	713	0.08	0.20	4.0E+03	0.0099	0.0009	0.08	0.00	38.74
627	2.39E-05	730	715	0.02	0.07	1.5E+03	0.0127	0.0002	0.01	0.00	52.81

Pipe type: 150 mm Radius Bend
 Internal Diameter (d) = 0.0202 m
 Area (A) = 0.0003205 m²
 Length (l) = 0.652 m

Time for 15 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Upstream Tapping (mm)	Downstream Tapping (mm)	Difference (Δh) (m)	Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
81	1.85E-04	899	681	0.22	0.58	1.2E+04	0.0076	0.0042	0.21	0.02	12.63
85	1.76E-04	878	679	0.20	0.55	1.1E+04	0.0077	0.0038	0.19	0.02	12.68
98	1.53E-04	834	679	0.15	0.48	9.6E+03	0.0080	0.0030	0.15	0.01	13.07
234	6.41E-05	755	678	0.07	0.20	4.0E+03	0.0099	0.0007	0.07	0.00	36.29
627	2.39E-05	701	683	0.02	0.07	1.5E+03	0.0127	0.0001	0.01	0.00	52.81

Pipe type: Mitre Corner
 Internal Diameter (d) = 0.0136 m
 Area (A) = 0.0001453 m²
 Length (l) = 0.94 m

Time for 15 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Upstream Tapping (mm)	Downstream Tapping (mm)	Difference (Δh) (m)	Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	u ² /2g	k _L
86	1.74E-04	1054	629	0.43	1.20	1.6E+04	0.0070	0.0355	0.39	0.07	5.78
92	1.63E-04	1006	640	0.37	1.12	1.5E+04	0.0071	0.0316	0.33	0.06	5.70
168	8.93E-05	942	653	0.29	0.61	8.3E+03	0.0083	0.0110	0.28	0.02	15.01
234	6.41E-05	824	672	0.15	0.44	6.0E+03	0.0090	0.0062	0.15	0.01	15.32
1150	1.30E-05	712	689	0.02	0.09	1.2E+03	0.0134	0.0004	0.02	0.00	55.97

Pipe type: Elbow Bend
 Internal Diameter (d) = 0.0136 m
 Area (A) = 0.0001453 m²
 Length (l) = 0.94 m

Time for 15 Liters (s)	Flow Rate (Q) (m ³ .s ⁻¹)	Upstream Tapping (mm)	Downstream Tapping (mm)	Difference (Δh) (m)	Flow Velocity (m.s ⁻¹)	Re	Blasius f	Straight Pipe Loss	Bend Loss	$u^2/2g$	k_L
86	1.74E-04	990	583	0.41	1.20	1.6E+04	0.0070	0.0355	0.37	0.07	5.54
92	1.63E-04	949	595	0.35	1.12	1.5E+04	0.0071	0.0316	0.32	0.06	5.51
168	8.93E-05	892	612	0.28	0.61	8.3E+03	0.0083	0.0110	0.27	0.02	14.54
234	6.41E-05	788	639	0.15	0.44	6.0E+03	0.0090	0.0062	0.14	0.01	15.01
1150	1.30E-05	692	665	0.03	0.09	1.2E+03	0.0134	0.0004	0.03	0.00	65.71